Question	Answer	Marks
5(a)	<u>sum of</u> e.m.f.(s) = <u>sum of</u> p.d.(s)	**M1**
	around a loop / around a closed circuit	**A1**
5(b)(i)	1 $6.0 - 4.0I = 0$ $I = 1.5$ A	**A1**
	2 $6.0 + 6.0 = I\,(4.0 + R + 1.5)$ $12 = 1.5\,(4.0 + R + 1.5)$	**C1**
	$R = 2.5\,\Omega$	**A1**
	or $6.0 = I\,(R + 1.5)$ $6.0 = 1.5\,(R + 1.5)$	**(C1)**
	$R = 2.5\,\Omega$	**(A1)**
	or combines $6 = 4I$ and $6 = I(R + 1.5)$ to give $4 = R + 1.5$	**(C1)**
	$R = 2.5\,\Omega$	**(A1)**

Question	Answer	Marks
5(b)(ii)	$I = Anvq$ ratio = $1^2 / 2^2$	**C1**
	= 0.25	**A1**
5(b)(iii)	<u>total</u> (circuit) resistance increases	**B1**
	current / I decreases or $P \propto I$ or $P \propto 1$ / (<u>total</u> resistance)	**M1**
	power (transformed) decreases	**A1**

Question	Answer	Marks
6(a)	1 / decreases by 1	**A1**